AI Usage in Research Project (IT4010)

Aims

- Responsible Use of AI Tools
- Student Orientation
- Understand • Justify • Own Your Work

defend

Why This Policy Matters

Report
Solution
Code Sequent
Repo.

problem → Solution

tool → output → brin

SSE

- AI is powerful—but risky if misused
- RP evaluates YOUR understanding, not AI output

→ understand
→ Analyze
→ Justify
→ defend

Professionals → 1

- Supports Learning Outcomes LO1–LO5

→ LO3 → tools.
→ LO4 → Tools

Core Principle

- AI is a SUPPORT tool, not a SUBSTITUTE
- You must understand everything you submit
- You are fully responsible for outcomes

→ Problem solving skills.

→ Critical thinking?

Understand.
Analyze - critically evaluate.
→ adapt
defend

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Where You CAN Use AI

- Learning concepts and theories
- Brainstorming ideas
- Debugging and coding support
- Improving writing clarity

Assistance can be taken from Any AI Tool

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Where You CANNOT Use AI

- Blind copy-paste from AI
- Generating full reports/proposals
- Using models without understanding
- Hiding AI usage

Academic Misconduct
Dishonesty.

Extent of AI Usage

- AI should assist—not dominate
- Core work must be YOUR thinking
- Design, justification, and implementation = YOUR responsibility

Mandatory AI Disclosure

Must include in all reports:

- Tools used
- Purpose
- Extent
- Validation method

What Panel Will Ask (Understanding)

- "Can you explain this part of your solution in your own words?"
- "How does this algorithm/model actually work?"
- "What happens internally when this code runs?"
- "Why does your system architecture follow this structure?"
- **What they are checking:**
- Do you actually understand the solution?
- Or did AI generate it for you? Engineer? or Slave?

What Panel Will Ask (Justification)

- "Why did you choose this model/technology?"
- "Why not use another approach?"
- "What are the limitations of your chosen method?"
- "How does this improve over existing solutions?"

This aligns with:

- Technology justification requirements
- Research gap and novelty expectations

What Panel Will Ask (AI Usage)

- "Did you use AI tools in this project?"
- "Where exactly did you use AI?"
- "What parts were generated vs developed by you?"
- "How did you validate AI-generated outputs?"

What they are checking:

- Honesty and academic integrity
- Responsible use of AI

What Panel Will Ask (Code Ownership)

- "Can you explain this function line by line?"
- "Why did you structure the code this way?"
- "What happens if this input changes?"
- "Can you modify this logic now?"

What they are checking:

- Did YOU write and understand the code?

What Panel Will Ask(Pre-trained Models)

- "Did you train this model or use a pre-trained one?"
- "What dataset was used?"
- "What are the biases or limitations?"
- "How did you adapt it to your problem?"

This aligns with:

- Data + ethical + evaluation expectations

What Panel Will Ask (Validation)

- “How did you verify your results?”
- “How do you know your solution works?”
- “What metrics did you use?”
- “What are the expected accuracy/performance levels?”

Why this matters:

- AI-generated outputs often look correct—but may be wrong.

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Pre-trained Models

Example project: “Plant disease detection”

Weak approach:

→ Use pretrained ResNet

→ Show prediction

Strong approach:

→ Compare multiple models

→ Fine-tune on the Sri Lankan crop dataset

→ Analyze misclassification cases

→ Integrate the system into the farmer advisory platform

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Consequences of Misuse

- Cannot explain → Lose marks
- Poor justification → Lower grades
- Misuse → Academic penalties

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Student Responsibility

YOU are accountable for:

- Code correctness
- Research accuracy
- Ethical implications
- Final results

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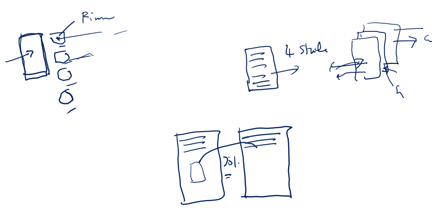
Final Message

- AI can help you finish
- Understanding helps you succeed
- Be the ENGINEER, not the COPY-PASTER



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